

Quiz 0 — Probability Review

Sep 4, 2025

Time limit: 20 minutes

Instructions: Show key steps for Q2, Q3, Q4.

Q1 [6 pts] Fill in the blanks:

1a. [1 pt] For any events A, B : $\Pr(A \cup B) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} - \underline{\hspace{1cm}}$.1b. [1 pt] Definition of conditional probability (for $\Pr(B) > 0$): $\Pr(A | B) = \underline{\hspace{1cm}}$.1c. [1 pt] Law of total probability: If $\{B_1, B_2, \dots, B_n\}$ is a partition of the sample space,

$$\Pr(A) = \sum_{i=1}^n \underline{\hspace{1cm}}.$$

1d. [1 pt] Bayes' rule (two events): $\Pr(B | A) = \underline{\hspace{1cm}}$.1e. [1 pt] The Law of Large Numbers implies that for i.i.d. $X_1, X_2, \dots, X_n$ with mean μ and variance σ^2 , the sample average $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ converges to $\underline{\hspace{1cm}}$ as $n \rightarrow \infty$.1f. Let $X \geq 0$ with $\mathbb{E}[X] = 5$. According to the Markov Inequality, $\Pr(X \geq 15) \leq \underline{\hspace{1cm}}$.

Q2 [6 pts] Consider the joint pmf:

$p_{X,Y}(x, y)$	$x = 0$	$x = 1$
$y = -1$	2/8	1/8
$y = 0$	1/8	2/8
$y = 1$	1/8	1/8

3a. [4 pts] Find $p_X(x)$ and $p_Y(y)$.3b. [2 pts] Prove that X and Y are not independent using one entry of the joint pmf.

Q3 [8 pts] Consider the joint pmf:

$p_{X,Y}(x, y)$	$x = 0$	$x = 1$
$y = 0$	1/4	1/4
$y = 1$	1/2	0

4a. [2 pts] Find $p_{X|Y}(x | 0)$.4b. [2 pts] Find $\mathbb{E}[X | Y = 0]$.4c. [2 pts] Find $p_{X|Y}(x | 1)$.4d. [2 pts] Let $W = \mathbb{E}[X | Y]$. Find the distribution of W .Q4 [2 bonus pts] In Problem Q2, find the pmf of $Z = \mathbb{E}[|Y| | X]$, where $|Y|$ is the absolute value of Y .